



DexterChem

REVISION SERIES ELECTROCHEMISTRY



NOTE

A dedicated Revision playlist link is in the description box of this chapter.

Containing :

- Revision Video***
- 2022 PYQs***
- 2023 PYQs***

To get Notes + PYQs of this chapter

Link in description or go to www.dexterchem.in/courses to find PYQ and Revision Notes.

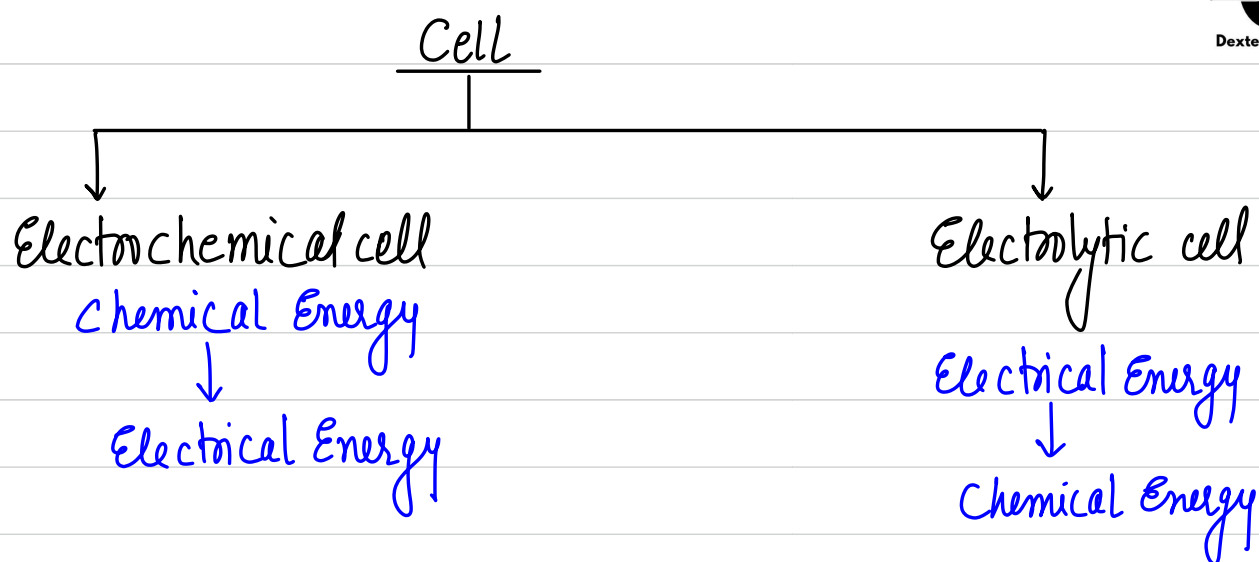
Electrochemistry

- ① Electrochemistry is the study of production of electricity from the energy released during a spontaneous reaction ($\Delta G < 0$) (Discharging)
- ② we use electrical energy to bring about non spontaneous reaction into spontaneous one.

Q: what is electrode?

Ans: when a metallic conductor is in association with electrolyte, the assembly is known as electrode or half cell.

cell: when two half cells are connected through a conductor then the assembly is known as a cell.



Electrochemical cell:-

- ① Chemical cell
- ② Concentration cell

① Chemical cell:-

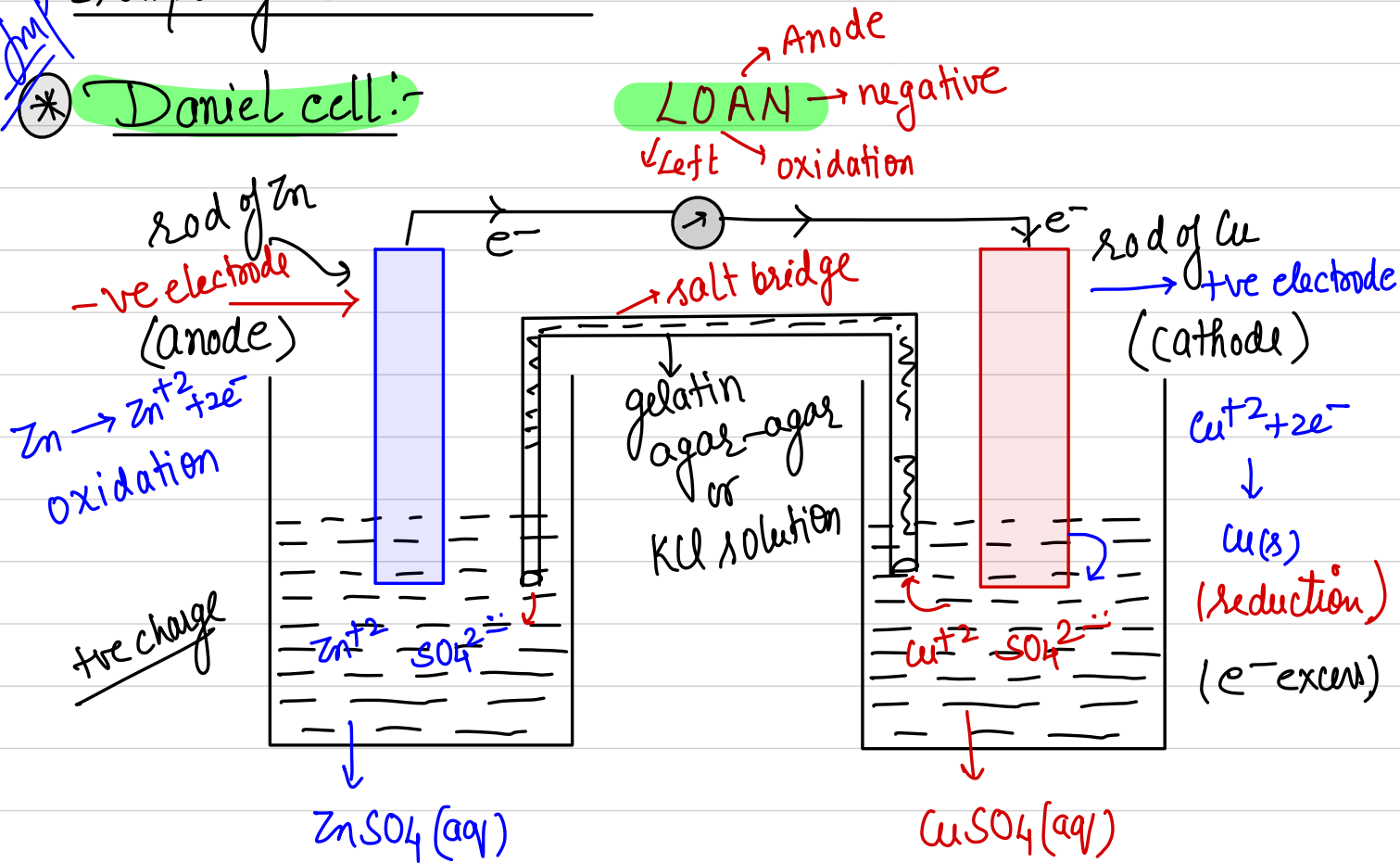
A net chemical change occurs inside the cell which is responsible for the flow of current.

② Concentration Cell:-

A net change in concentration is found inside the cell which is responsible for the flow of current.

Example of Chemical Cell:-

* Daniel cell:-



AO CR
 ↓ ↓
 Anode oxidation Cathode reduction

* loss of e^- s is called oxidation.

* Gain of e^- s is called reduction.

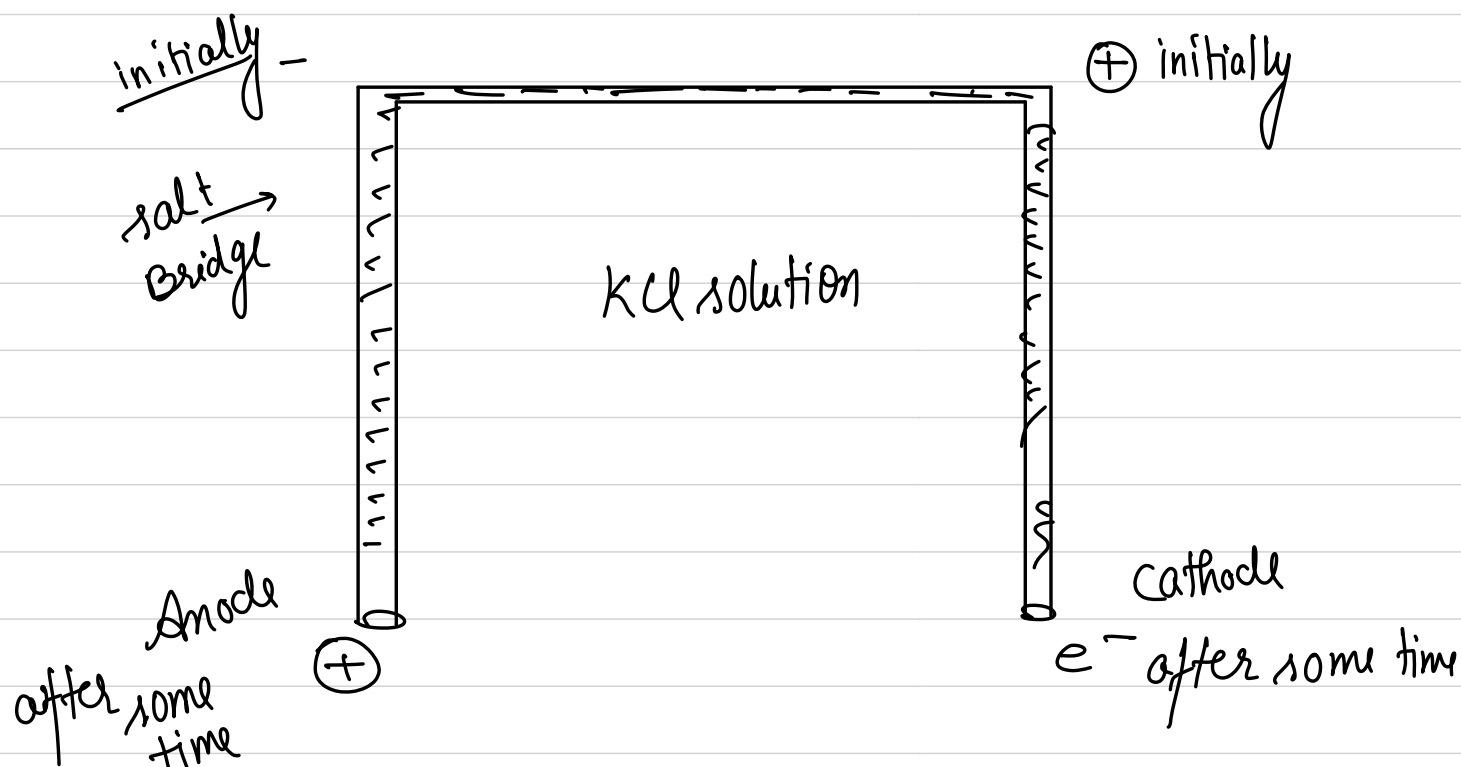
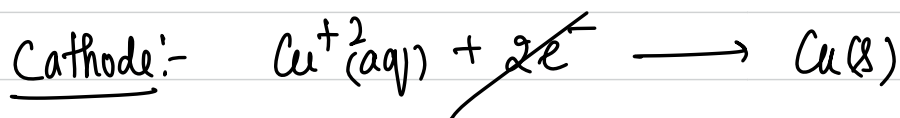
Anode:- oxidation



Cathode:- reduction



Overall rxn:-



function of Salt Bridge:-

* The main function

of salt bridge is to maintain the electrical neutrality.

- * It completes the circuit by allowing flow of e^- s.
- * It minimizes the liquid junction potential.

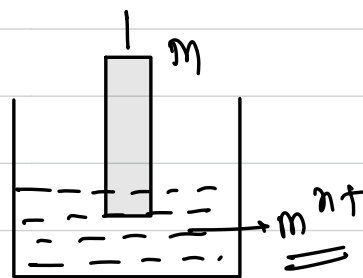
Characteristics of the electrolyte used in salt bridge.

① ~~Imp~~ The electrolyte should not react with the electrolyte of cathode or anode.

② The ionic mobility of cation & anion of the salt bridge should be nearly same.

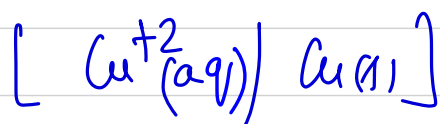
Types of half cells:-

① Metal-metal ion half cell:-



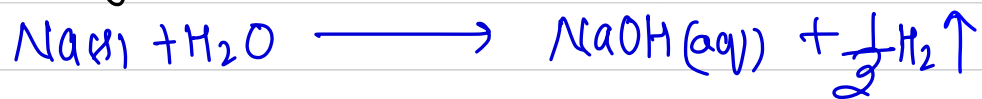
In this the metallic rod is immersed in a solution containing its own ions.

eg. CuSO_4 solution containing Cu-electrode



* the elements involved in the half cell are generally from the electrode.

* We can not use very reactive metals because they generally react with H_2O .

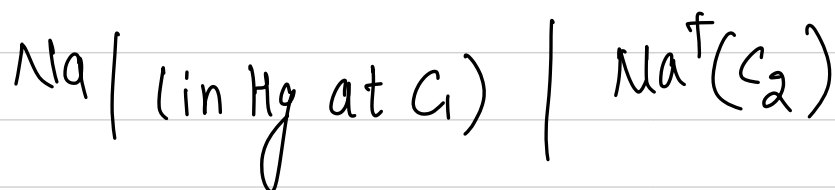


Imp * We can not prepare $Na^+(aq)/Na$ electrode.

Amalgam electrode:- The metal is in the form of an amalgam i.e. it is dissolved / dispersed in mercury rather than in pure form.

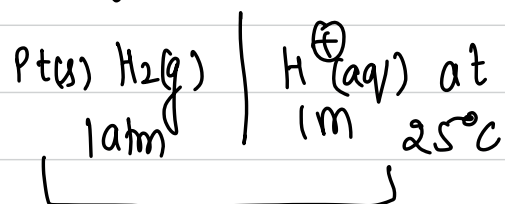
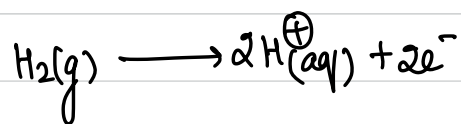
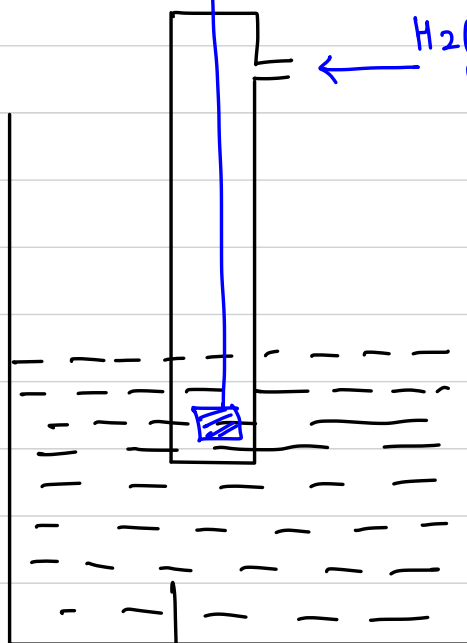
The electrical contact is made by a Pt wire dipping into amalgam pool.

The rxn is same as metal/metal ion electrode with Hg playing no role.

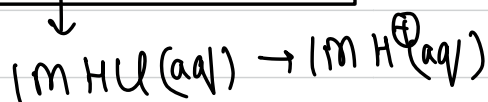


Gas-Gas ion electrode:-

ignition → Pt wire inert electrode.
 $H_2(g)$ or $X(g)$

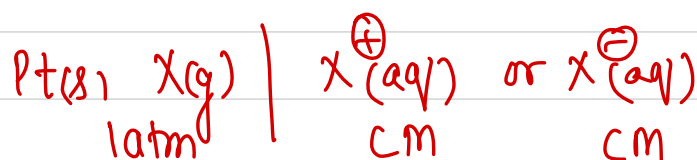


SHE → Standard Hydrogen Electrode.



$$E^\circ_{\text{cell}} = 0 \text{ V}$$

In general



Gas

H_2

Cl_2

O_2

SO_2

CO

Gas Ion

H^+

Cl^-

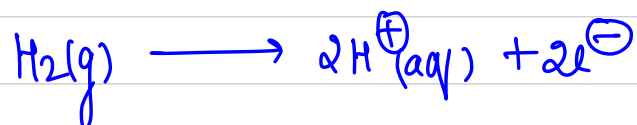
$\ddot{O}H(aq) | H_2O(l)$

$SO_4^{2-}(aq)$

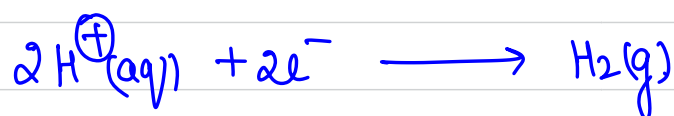
CO_3^{2-} / HCO_3^-

(a) For hydrogen Electrode:-

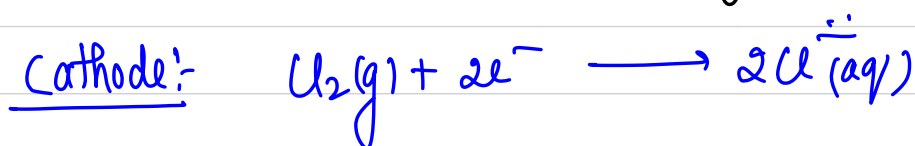
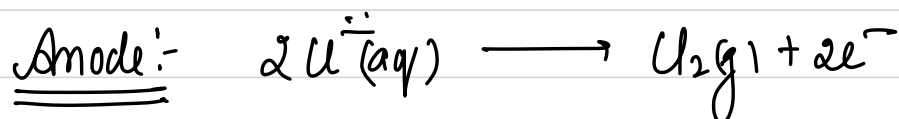
(i) Oxidation half cell (Anode)



(ii) Reduction half cell (Cathode)

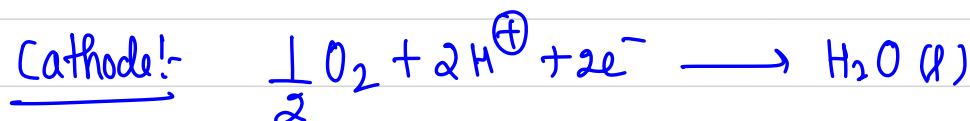
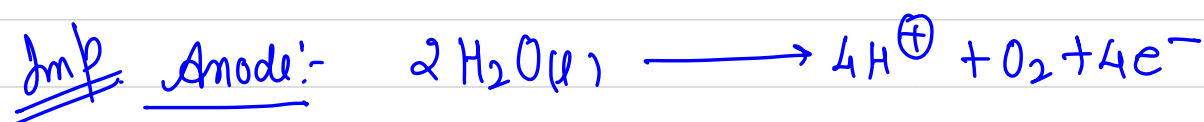


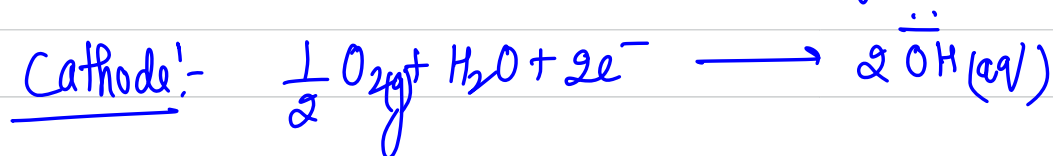
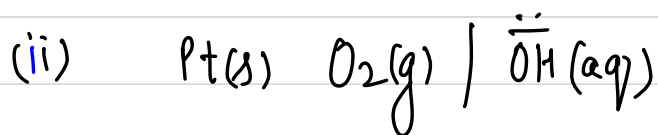
(b) For chlorine electrode:-



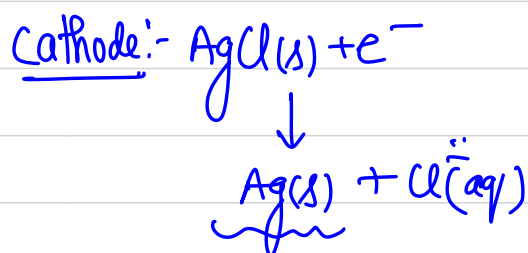
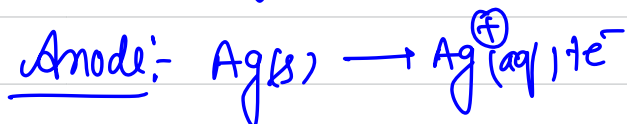
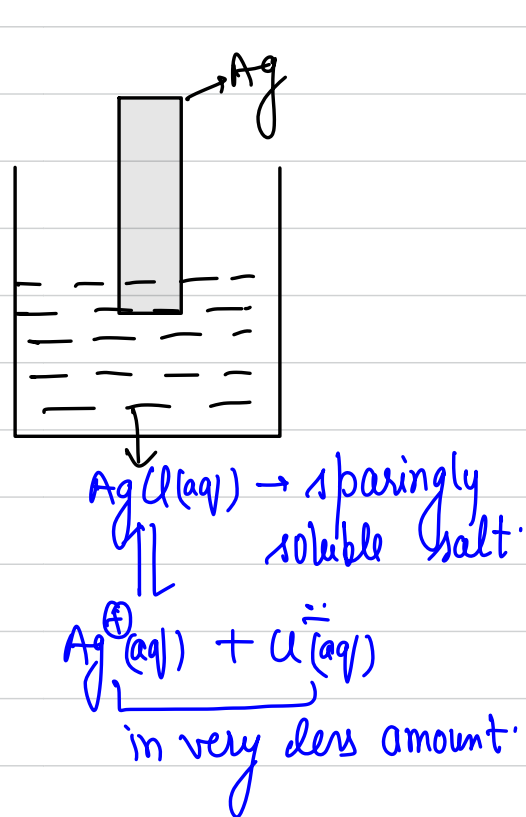
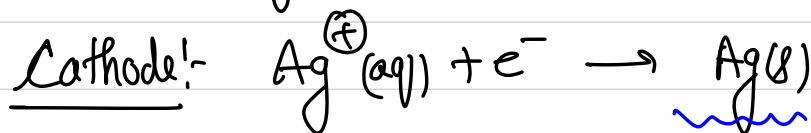
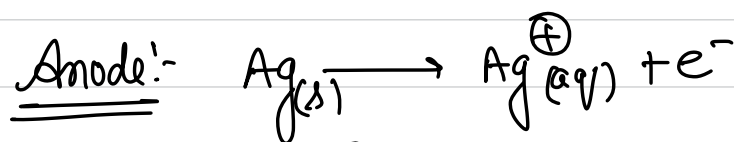
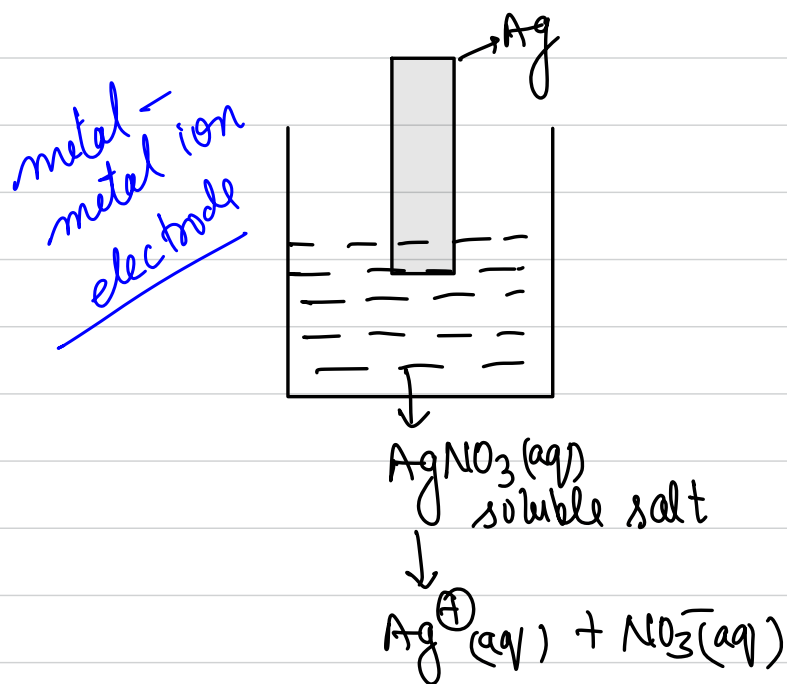
(c) Oxygen Electrode:-

(i) $\text{Pt(s)} \mid \text{O}_2(\text{g}) / \text{H}_2\text{O(l)}$





* metal-metal ion insoluble salt Electrode:-



Redox electrode:-

In this a platinum wire is placed in a solution of electrolyte containing ions of an element in more than one oxidation state.

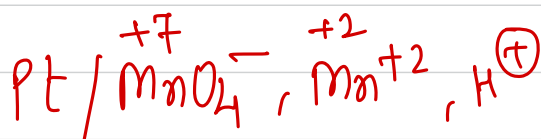
for eg:- Pt wire is dipped in an electrolyte containing Fe^{+2} & Fe^{+3} .

Electrode representation : $Pt | Fe^{+2}, Fe^{+3}$

electrode rxn:-



eg:-



Electrode potential:-

It is impossible to measure the absolute EMF of a half cell. We can calculate the relative EMF w.r.t some other half cell.

* The reference electrode against which all other half cell are generally measured is SHE.

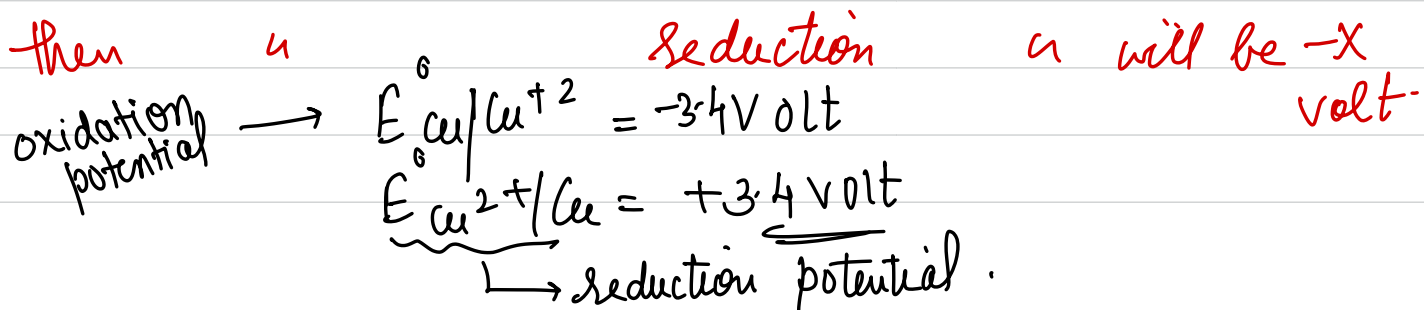
$$E_{SHE}^{\circ} = \underline{\underline{0.0V}}$$

How to assign the sign of electrode potential:-

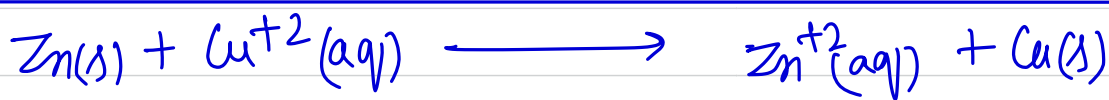
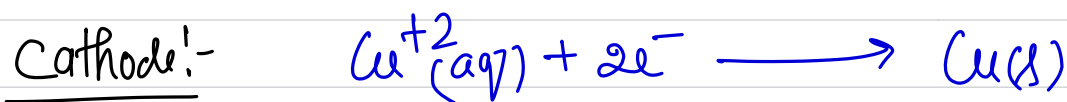
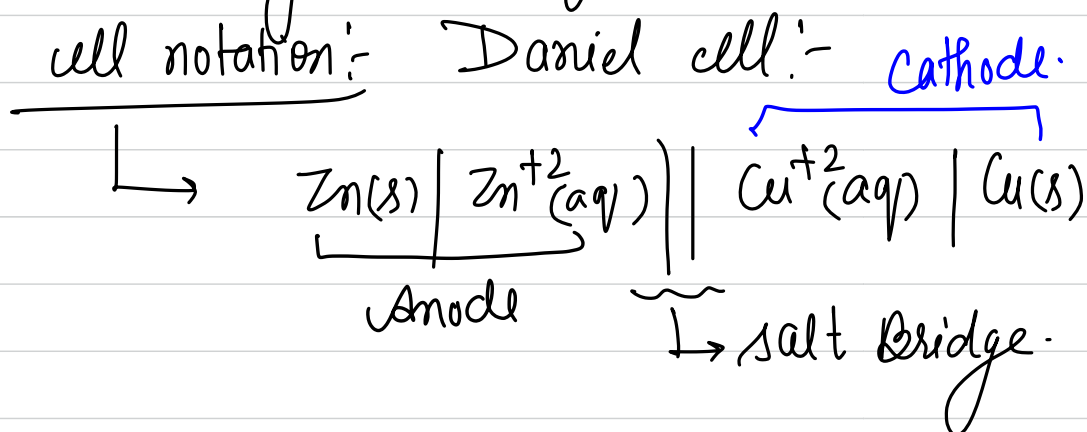
The oxidation potential of the half cell is given a +ve sign. If the electrode acts as the anode in the cell.

Reduction potential is given a -ve sign.

Let us suppose that the oxidation potential is x volt.



Representation of a cell diagram:-



The spontaneous chemical rxn is responsible for the net decrease in free energy.

$$W = \text{Charge} \times \text{potential}$$

$$W = q \times E$$

$$-\Delta G = W$$

$$-\Delta G = nFE$$

$$\Delta G = -nFE$$

$$\Delta G^{\circ} = -nFE^{\circ}$$

moles of charge.
↓
n-factor
↓
Charge = $n \times F$ Faraday.

$q = \underline{nF} \rightarrow n\text{-Faraday charge.}$

$$1F = 96500C$$

$$1F = \text{charge of } 1\text{mol of } e^-$$

$$= 1.6 \times 10^{-19} \times 6.023 \times 10^{23} \text{ C}$$

$$1F = 96500 \text{ C}$$

$$\Delta G = -nFE$$

$$\Delta G^\circ = -nFE^\circ$$

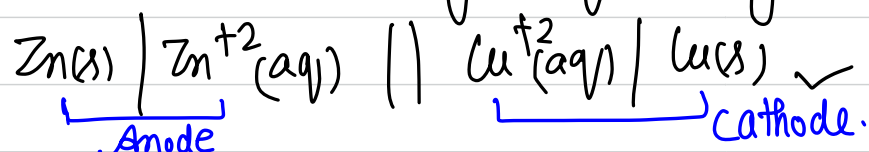
Calculation of E° :-

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$$

$$E_{\text{cell}} = E_C - E_A$$

E_C or E_A } should always be reduction potential.

Q: Calculate the E°_{cell} of the following cell diagram.



given $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76\text{V}$

reduction potential. $\leftarrow E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34\text{V}$

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$= E^\circ_{\text{Cu}^{2+}/\text{Cu}} - E^\circ_{\text{Zn}^{2+}/\text{Zn}}$$

$$= +0.34 - (-0.76\text{V})$$

$$= \underline{\underline{+1.10\text{V}}}$$

* E_{cell} is an intensive property & can not be additive.
 ↓
 * are there which do not depend upon the mass. eg: density.

* ΔG° is an extensive property (additive in nature)
 ↓
 * are there properties which depend upon the mass.

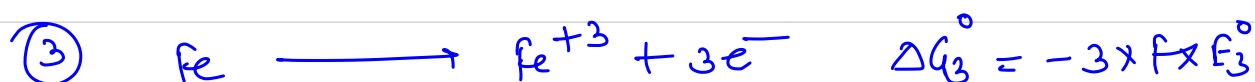
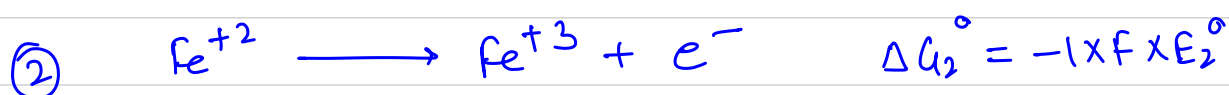
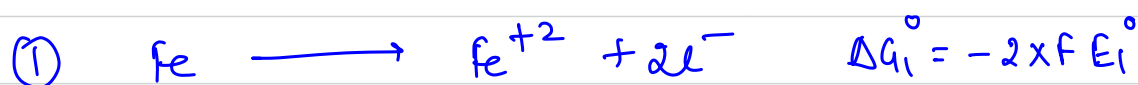
~~* bnp~~ E_{cell} of a cell depends upon the make up of the cell where as ΔG° depends only upon the net work done by the cell. & not on make up of the cell.

~~bnp~~ (*) We can directly add or subtract E° if and only if the n-factors in the rxns are same.

Latimer Diagram:

Q: If E_1° is the standard electrode potential for Fe/Fe^{+2}
 E_2° is for $\text{Fe}^{+2}/\text{Fe}^{+3}$ & E_3° is for Fe/Fe^{+3}

Derive relation b/w E_1° , E_2° & E_3° .



$$\textcircled{3} = \textcircled{1} + \textcircled{2}$$

$$\times \left[E_3^\circ = E_1^\circ + E_2^\circ \right] \Rightarrow \text{wrong}$$

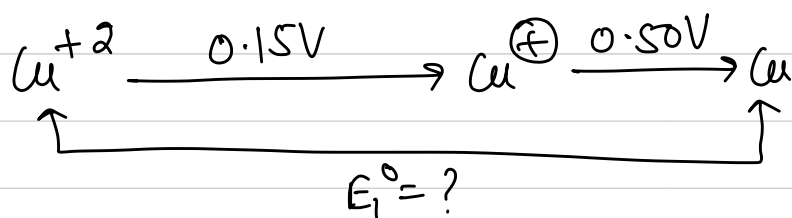
$$\Delta G_3^\circ = \Delta G_1^\circ + \Delta G_2^\circ$$

$$\cancel{-3 \times F \times E_3^\circ} = \cancel{-2 \times F \times E_1^\circ} - \cancel{1 \times F \times E_2^\circ}$$

$$3 E_3^\circ = 2 E_1^\circ + E_2^\circ$$

$$E_3^\circ = \frac{2 E_1^\circ + E_2^\circ}{3}$$

HW
Q: From the standard potential shown in the following diagram. Calculate the value of E_1° ?



(NERNST EQUATION)

Relation b/w the EMF & equilibrium constant of a cell rxn:-

$$\# \Delta G = \Delta G^\circ + RT \ln Q$$

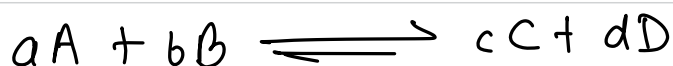
$$-nFE = -nFE^\circ + RT \ln Q$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$$

$$\left. \begin{array}{l} T = 298\text{K} \\ F = 96500 \\ R = 8.314 \end{array} \right\}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n} \log Q$$

$Q \rightarrow$ Rxn Quotient $\Rightarrow$



$$Q = \frac{[C]^c [D]^d}{\underbrace{[A]^a [B]^b}} = \frac{p_C^c \times p_D^d}{\underline{\underline{p_A^a \times p_B^b}}}$$

at Equilibrium $Q = K \rightarrow$ Equilibrium constant.

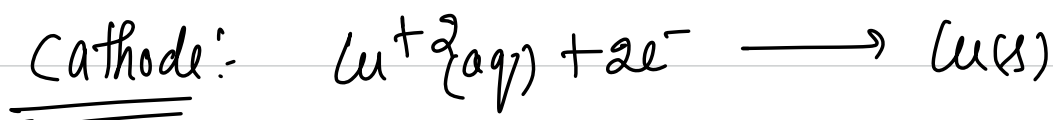
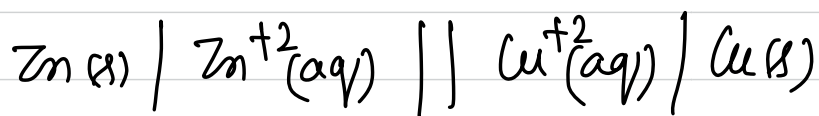
$\downarrow$

$$\Delta G = 0$$

$$\Delta G^\circ = RT \ln K$$

$$E^\circ_{\text{cell}} = \frac{+0.0591}{n} \log K$$

for Daniel cell:-



$$Q = \frac{[\text{Zn}^{+2}(\text{aq})]}{[\text{Cu}^{+2}(\text{aq})]} = \frac{(\text{Anode})}{(\text{Cathode})}$$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Zn}^{+2}(\text{aq})]}{[\text{Cu}^{+2}(\text{aq})]}$$

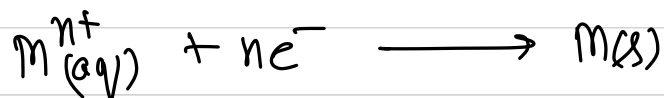
$$E_{\text{cell}} = [E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}] - \frac{0.0591}{2} \log \frac{[\text{Zn}^{+2}]}{[\text{Cu}^{+2}]}$$

in general

$$E_{\text{cell}} = E^{\circ}_{\text{C}} - E^{\circ}_{\text{A}} - \frac{0.0591}{n} \log \frac{(\text{Anode})}{(\text{Cathode})}$$

Application of NERNST Equation:-

① Emf of an electrode (half cell)



$$Q = \frac{1}{[M^{n+}]}$$

$$E_{cell} = E_{M^{n+}/M}^{\circ} - \frac{0.0591}{n} \log \frac{1}{[M^{n+}]}$$

② Calculation of pH:-



$$p_{H_2(g)} = \underline{\underline{1 \text{ atm.}}}$$

$$E_{cell} = E_{H^{+}/H_2}^{\circ} - \frac{0.0591}{1} \log \frac{p_{H_2}}{[H^{+}]}$$

$$= 0 - \frac{0.0591}{1} \log \frac{1}{[H^{+}]}$$

$$E_{cell} = -0.0591 (-\log [H^{+}])$$

$$E_{cell} = -0.0591 \text{ pH}$$

Imp Temp. dependence of EMF of the cell:-

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta S = - \left(\frac{\partial(\Delta G)}{\partial T} \right)_P$$

$$\Delta G = \Delta H + T \left(\frac{\partial(\Delta G)}{\partial T} \right)_P$$

$$-nFE_{cell} = \Delta H + T \left(\frac{\partial(-nFE_{cell})}{\partial T} \right)_P$$

$$-nFE_{cell} = \Delta H - nFT \left(\frac{\partial E_{cell}}{\partial T} \right)_P$$

Imp

$$\left(\frac{\partial E_{cell}}{\partial T} \right)_P = \frac{E_{cell}}{T} + \frac{\Delta H}{nFT}$$

Use of Nernst equation in determining K_{sp} of a

sparingly soluble salt:-

partially soluble $\rightarrow$ AgCl

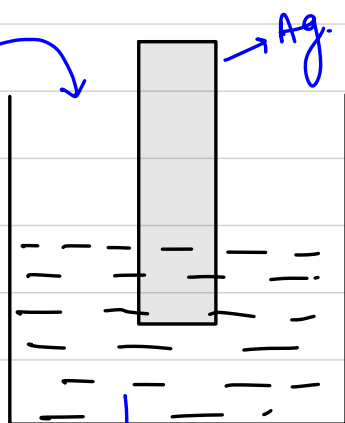
*

$$E_{Cl^-/AgCl/Ag}^{\circ} = E_{Ag^+/Ag}^{\circ} + \frac{0.0591}{1} \log K_{sp}$$

metal insoluble salt electrode

solubility product

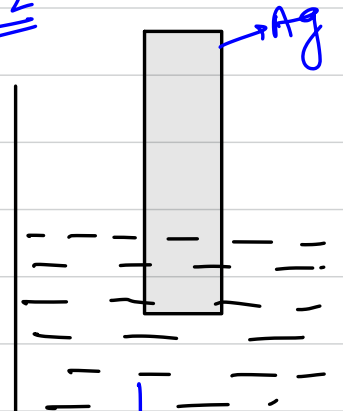
Case-1
excess NaCl



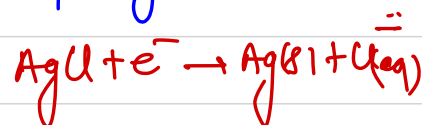
AgNO₃(aq) $\rightarrow$ soluble.



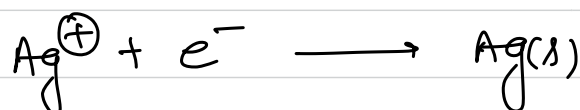
Case-2



AgCl(aq)
sparingly soluble.



Case-1



$$E_{cell} = E_{Ag^+/Ag}^{\circ} - \frac{0.0591}{1} \log \frac{1}{[Ag^+]}$$

Now multiply & divide $[Cl^-]$ in the log

$$E_{\text{cell}} = E^{\circ}_{\text{Ag}^+/\text{Ag}} - \frac{0.0591}{1} \log \frac{[\text{Cl}^-]}{[\text{Cl}^-][\text{Ag}^+]}$$

$$K_{\text{sp AgCl}} = [\text{Ag}^+][\text{Cl}^-]$$

$$\textcircled{1} \quad E_{\text{cell}} = E^{\circ}_{\text{Ag}^+/\text{Ag}} - \frac{0.0591}{1} \log [\text{Cl}^-] + \frac{0.0591}{1} \log K_{\text{sp}}$$

Case 2



$$E_{\text{cell}} = E^{\circ}_{\text{Cl}^-/\text{AgCl}/\text{Ag}} - \frac{0.0591}{1} \log [\text{Cl}^-] \quad \text{---} \textcircled{2}$$

from both the cases the E_{cell} should be same
 $\textcircled{1} = \textcircled{2}$

$$E^{\circ}_{\text{Cl}^-/\text{AgCl}/\text{Ag}} - \frac{0.0591}{1} \log [\text{Cl}^-] = E^{\circ}_{\text{Ag}^+/\text{Ag}} - \frac{0.0591}{1} \log [\text{Cl}^-] + \frac{0.0591}{1} \log K_{\text{sp}}$$



$$E^{\circ}_{\text{Cl}^-/\text{AgCl}/\text{Ag}} = E^{\circ}_{\text{Ag}^+/\text{Ag}} + \frac{0.0591}{1} \log K_{\text{sp}}$$

HW

Q: If $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.8\text{V}$ and $E^\circ_{\text{H}^+/\text{H}_2} = 0\text{V}$

In a cell arrangement using these two electrodes which will act as cathode & anode.

Calculate the E°_{cell} of the spontaneous rxn.

Let Ag^+ is cathode ✓ & H^+/H_2 anode.

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$= 0.8 - 0.0$$

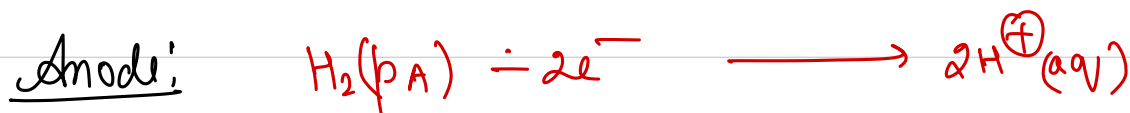
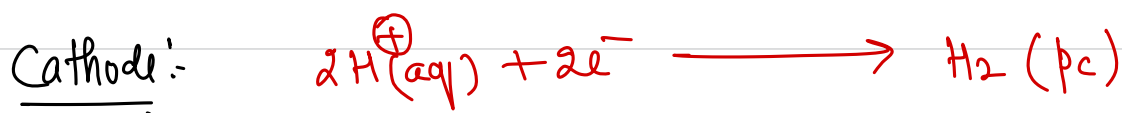
$$= \underline{\underline{0.8\text{V}}}$$

Concentration cell:- Due to the change in conc. of the cations & metals at different electrodes the EMF is generated.

- ① Electrode conc. cell
- ② Electrolyte conc. cell.

① Electrode conc. cell:- In these cells two like electrodes at different conc. are dipped in same solution.

Two hydrogen electrodes at unequal gas pressure are immersed in same solution of H^+ ions constitute an electrode conc. cell.



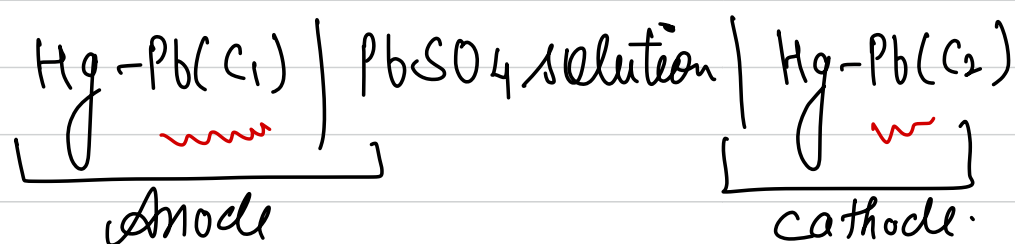
$$H_2(p_A) \longrightarrow H_2(p_c)$$

$$E_{cell} = E_{cell}^{\circ} - \frac{0.0591}{2} \log \left(\frac{p_c}{p_A} \right)$$

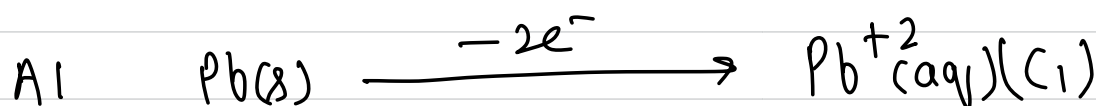
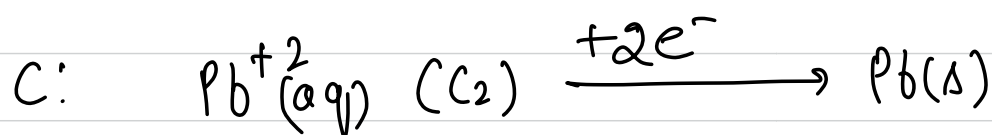
$\downarrow$
0

$$E_{cell} = \frac{0.0591}{2} \log \left(\frac{p_A}{p_C} \right)$$

Electrolyte conc. Cell:-



Rxn involved:-



$$E_{cell} = E_{cell}^{\circ} - \frac{0.0591}{2} \log \frac{[\text{Pb}^{+2}_{(aq)}(C_1)]}{[\text{Pb}^{+2}_{(aq)}(C_2)]}$$

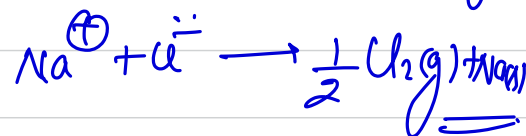
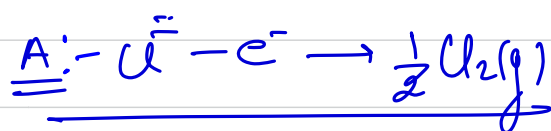
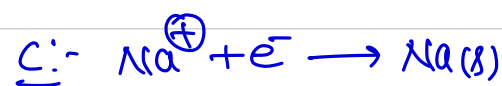
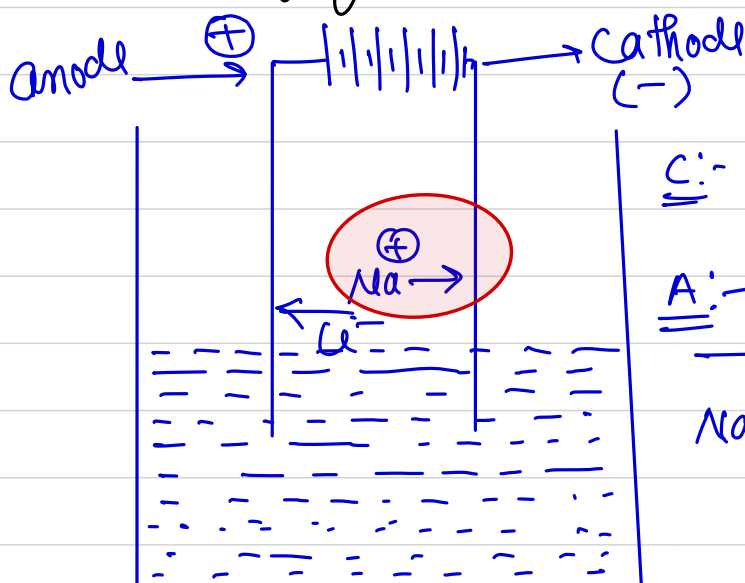
$\Downarrow$
0

$$E_{cell} = \frac{0.0591}{2} \log \frac{[\text{Pb}^{+2}_{(aq)}(C_2)]}{[\text{Pb}^{+2}_{(aq)}(C_1)]}$$

Electrolysis

During electrolysis, Electrical Energy is converted into chemical Energy.

Care-1 Electrolysis of fused NaCl (molten NaCl):-



Care-2 Electrolysis of pure water using inert electrode:-

for pure water $[\text{H}^+] = [\text{OH}^-] = 10^{-7} \text{M}$. at 25°C

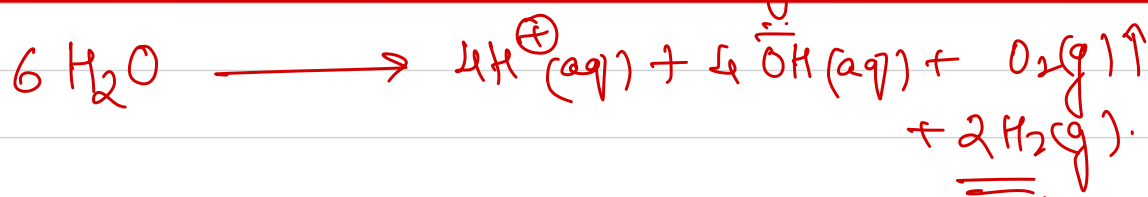
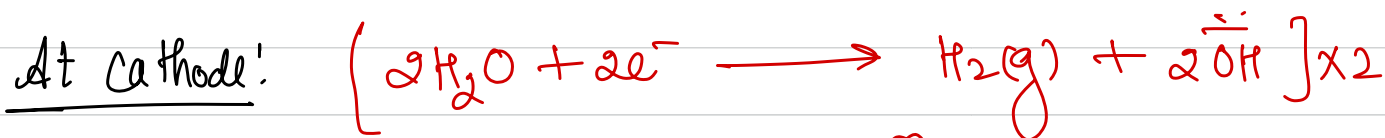
In comparison to H_2O the conc. of $[\text{H}^+]$ & $[\text{OH}^-]$ is very low.

$$[\text{H}_2\text{O}] = \frac{55.55 \text{M}}{18} = 1000/18$$

$$K_w \text{ of } \text{H}_2\text{O} \text{ at } 25^\circ\text{C} = 10^{-14}$$

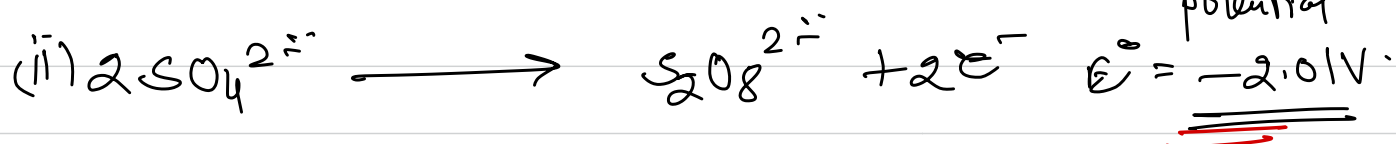
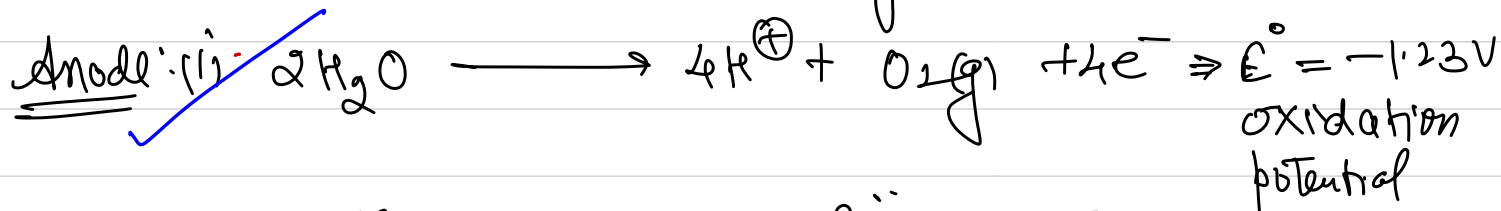
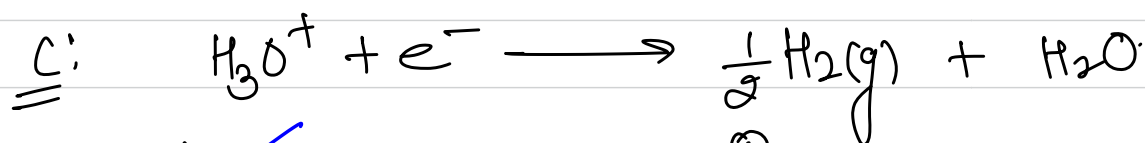
$$[\text{H}^+] = [\text{OH}^-] = 10^{-7} \text{M}$$

Therefore when the current is passed in water. The following reactions will occur.



Case-3 Electrolysis of acidified water (dil H_2SO_4):-

In case of acidified water, the amount of H_3O^+ is sufficient, then the oxidation of H_2O will take place & reduction of H_3O^+ will take place.



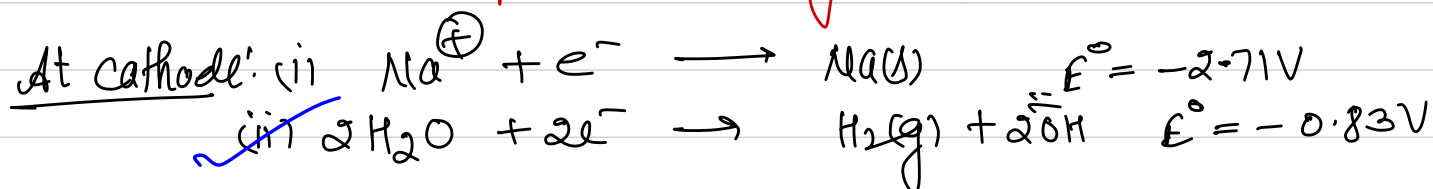
Since the oxidation potential & conc. of H_2O is more than SO_4^{2-} then the current passed will more effectively oxidize H_2O in comparison to SO_4^{2-} .

* In case the conc. of H_2SO_4 is large then oxidation of SO_4^{2-} will take place. (large > 38%)

Lead Storage Battery

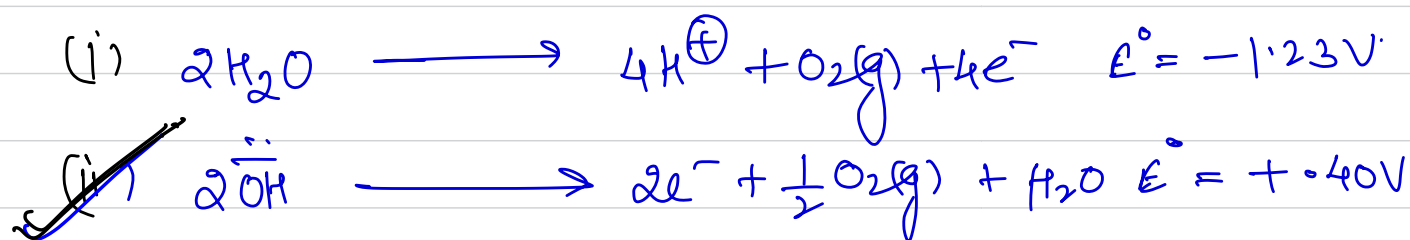
④ Electrolysis of alkaline water (dil NaOH):-

In this electrolysis there are two possible reactions that can take place at every electrode:-



In this case Na^+ ions are heavily hydrated & its reduction potential is also very less than H_2O .

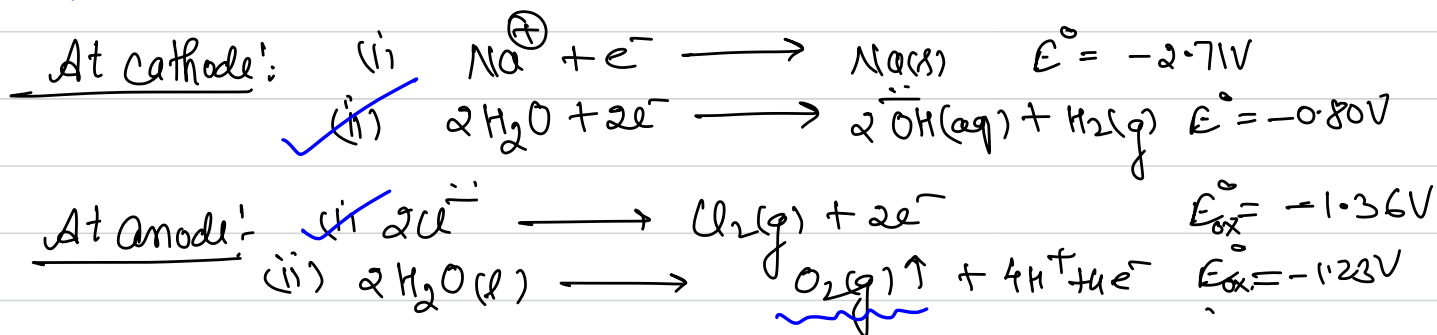
Similarly at Anode:-



~~imp~~ ~~imp~~ * The species whose oxidation potential will be higher with good conc. will preferably oxidized at anode. & the species whose reduction potential will be high with good conc. will preferably get reduced at cathode.

⑤ Electrolysis of aq. NaCl (Brine solution)

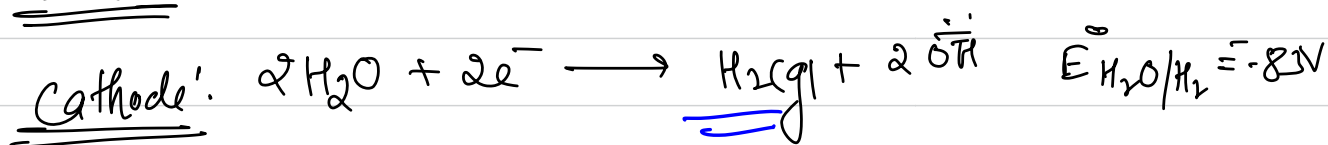
Available ions in the aq. solution are Na^+ , Cl^- , $\text{H}^+(\text{aq})$ & $\text{OH}^-(\text{aq})$.



over potential: It is the extra potential required to initiate a reaction at required rate.

* If E_{cell} for a reaction is less than 0.6V then it will be very slow kinetically.

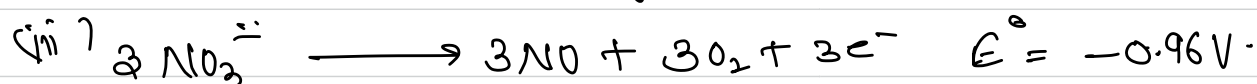
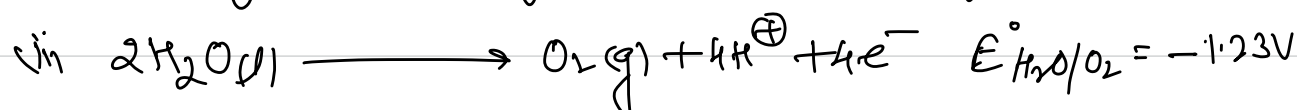
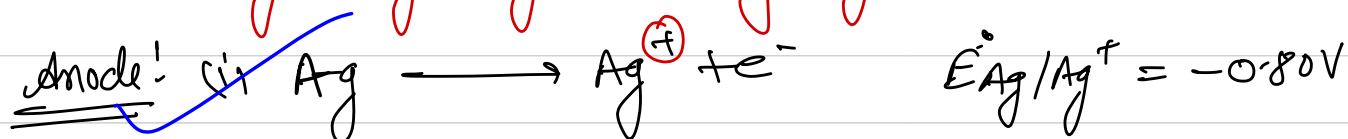
* Rxn of sodium with water.



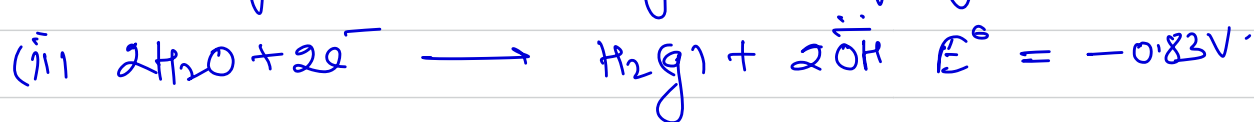
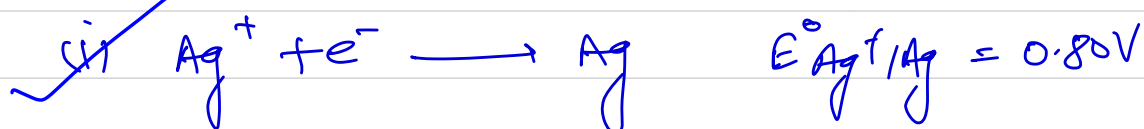
$$\begin{aligned}
 E^\circ_{\text{cell}} &= E^\circ_{\text{c}} - E^\circ_{\text{a}} \\
 &= -0.83 - (-2.71\text{V}) \\
 &= \underline{\underline{+1.88\text{V}}} > \underline{\underline{0.6\text{V}}}
 \end{aligned}$$

Electrolysis of solution in presence of reactive electrode:-

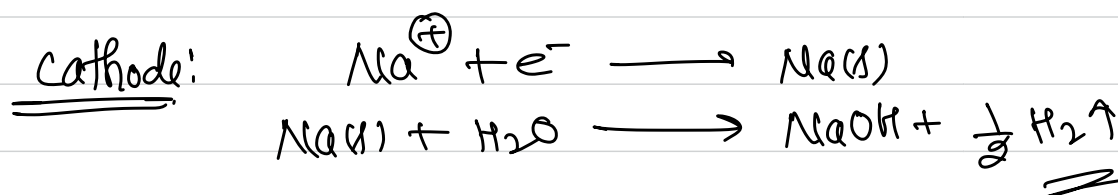
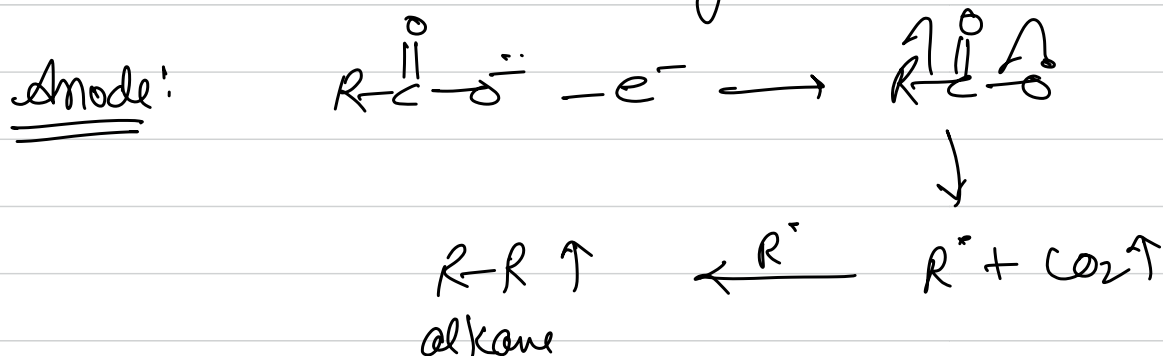
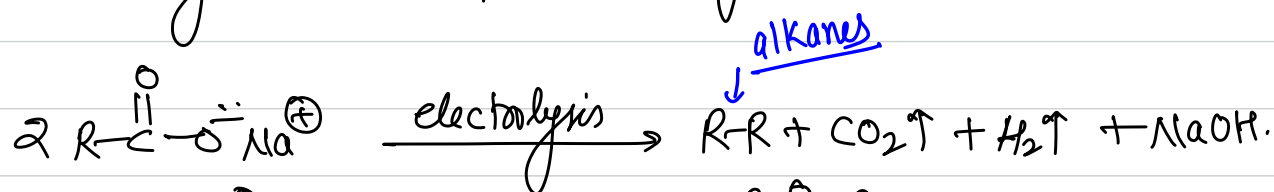
(a) Electrolysis of AgNO_3 using Ag electrode:-



Cathode:



Kolbe's Electrolysis:- (Aq. solution of $R-\overset{\overset{O}{\parallel}}{C}-\ddot{O}Na^+$)



As the reaction proceeds the pH of cathodic compartment increases & $H_2(g)$ gas is liberated at cathode.

At anode $CO_2(g)$ is liberated with alkanes.

Faraday's Law of Electrolysis:-

Ist Law :-

The amount of any substance liberated or deposited at any electrode is directly proportional to the quantity of charge passed through it.

$$w \propto Q$$

$Q \rightarrow$ charge passed.

$w \Rightarrow$ amount deposited

$$w \propto it$$

$$w = z it$$

$i =$ current
 $t =$ time

where z is called the electrochemical equivalent of the substance.

$$z = \frac{E}{96500}$$

$E \rightarrow$ Equivalent mass

$$1 \text{ Faraday} = 96500 \text{ C}$$

* The electrochemical equivalent is that mass of a substance which is liberated or deposited when 1 amp current is passed through the solution for 1 sec. (1C charge)

$$w = Z \quad \text{when } i = 1 \text{ amp}$$

$$Q = 1 \text{ C}$$

$$t = \underline{\underline{1 \text{ sec}}}$$

Faraday's Second Law:-

The mass of a substance deposited or liberated at any electrode on passing a certain amount of current is directly proportional to its chemical equivalent weight.

$$W \propto E$$

$$W = \frac{Q \times E}{96500}$$

$$\boxed{\frac{W_1}{W_2} = \frac{E_1}{E_2}}$$

$$\text{Equivalent mass} = \frac{\text{molar mass}}{n \text{ factor}}$$

Resistance:-

It is the hindrance provided by a conductor in the passage of current. It is measured in terms of Ω .

$$R \propto \frac{l}{a}$$

$l \rightarrow$ length

$a \Rightarrow$ area of cross-section

$$R = \rho \times \frac{l}{a}$$

$\rho \rightarrow$ Resistivity

$$\rho = R$$

$\Rightarrow$ for $l = 1 \text{ cm}$
 $a = 1 \text{ cm}^2$

Resistivity (ρ) is the resistance offered by a conductor wire having unit length & unit cross-sectional area.

Conductance:-

The ease with which the electricity flows through a solution.

$$G = \frac{1}{R}$$

unit = Ω^{-1} ohm $^{-1}$ or mho.

SI unit of conductance is Simen (S).

specific conductance (conductivity):-

$$k = \frac{1}{\rho}$$

$k \Rightarrow$ kappa

$$k = \frac{1}{R \times \left(\frac{a}{l}\right)}$$

$$k = \frac{1}{R} \times \frac{l}{a}$$

$$k = G \times \frac{l}{a}$$

$\left(\frac{l}{a}\right) \Rightarrow$ cell const.

$l \rightarrow$ distance b/w two electrodes

$a \rightarrow$ area of cross section.

unit of $k = \Omega^{-1} \times \frac{\text{cm}}{\text{cm}^2}$

$= \Omega^{-1} \text{cm}^{-1} \quad \text{or} \Rightarrow \underline{\underline{\text{S m}^{-1}}}$

Conductivity in solution

Equivalent conductance

molar conductance.

Equivalent conductance:-

The conducting power of all the ions produced by one gram equivalent of an electrolyte in a given solution. It is represented by Λ_{eq} .

Relationship b/w the specific conductance & equivalent

conductance :-

Let one gram equivalent of an electrolyte is dissolved in v ml of solution. Then all the ions produced by 1g equivalent of the electrolyte will be present in v ml of solution.

$$\Lambda_{eq} = \text{conductance of (1 cc) solution} \times \text{volume in cc.}$$

$$\Lambda_{eq} = k \times V$$

$$N = \frac{\text{no of } g \text{ equivalent}}{V} \times 1000$$

$$V = \frac{1}{N} \times 1000$$

~~am~~

$$\Lambda_{eq} = \frac{k \times 1000}{N}$$

$$\begin{aligned} \text{unit} &\Rightarrow \Omega^{-1} \text{cm}^{-1} \times \text{cm}^3 / \text{eq} \\ &= \Omega^{-1} \text{cm}^2 / \text{eq} \\ &= \underline{\underline{\text{Sm}^2 / \text{eq}}} \end{aligned}$$

Molar conductance:- Λ_m :-

It is defined as the conducting power of all the ions produced by 1 mole of electrolyte in given solution.

$$\Lambda_m = \frac{k \times 1000}{m}$$

$$\text{unit} \rightarrow \text{Sm}^2 / \text{mol}$$

$$= \Omega^{-1} \text{cm}^2 / \text{mol}$$

for sparingly soluble salt

$$\Lambda_m = \frac{k \times 1000}{s}$$

$s \rightarrow$ solubility

Variation of conductance with dilution:-

On lowering the conc. equivalent conductance or molar conductance increases whereas specific conductance decreases.

$$(i) \quad \Lambda_m \text{ or } \Lambda_{eq} \propto \frac{1}{[\text{conc.}]}$$

(ii) On increasing dilution ionic mobility increases & steric hindrance decreases.

* On increasing dilution, the no. of ions per unit volume decreases & therefore specific conductance decreases.

* At infinite dilution or almost zero conc. equivalent conductance & molar conductance attain their limiting max. value. This is called equivalent conductance or molar conductance at infinite dilution or zero conc.

$$\Lambda_{eq}^{\infty} \text{ or } \Lambda_m^{\infty}$$

$$\Lambda_{eq}^0 \text{ or } \Lambda_m^0$$

strong electrolytes:- They dissociate completely at every conc.

So increase in conductance is due to ionic mobility & decrease in inter-ionic attraction forces whereas in case of weak electrolytes, the increase in conductance is due to increased degree of dissociation in addition to increase in ionic mobility & decrease in inter-ionic attraction forces.

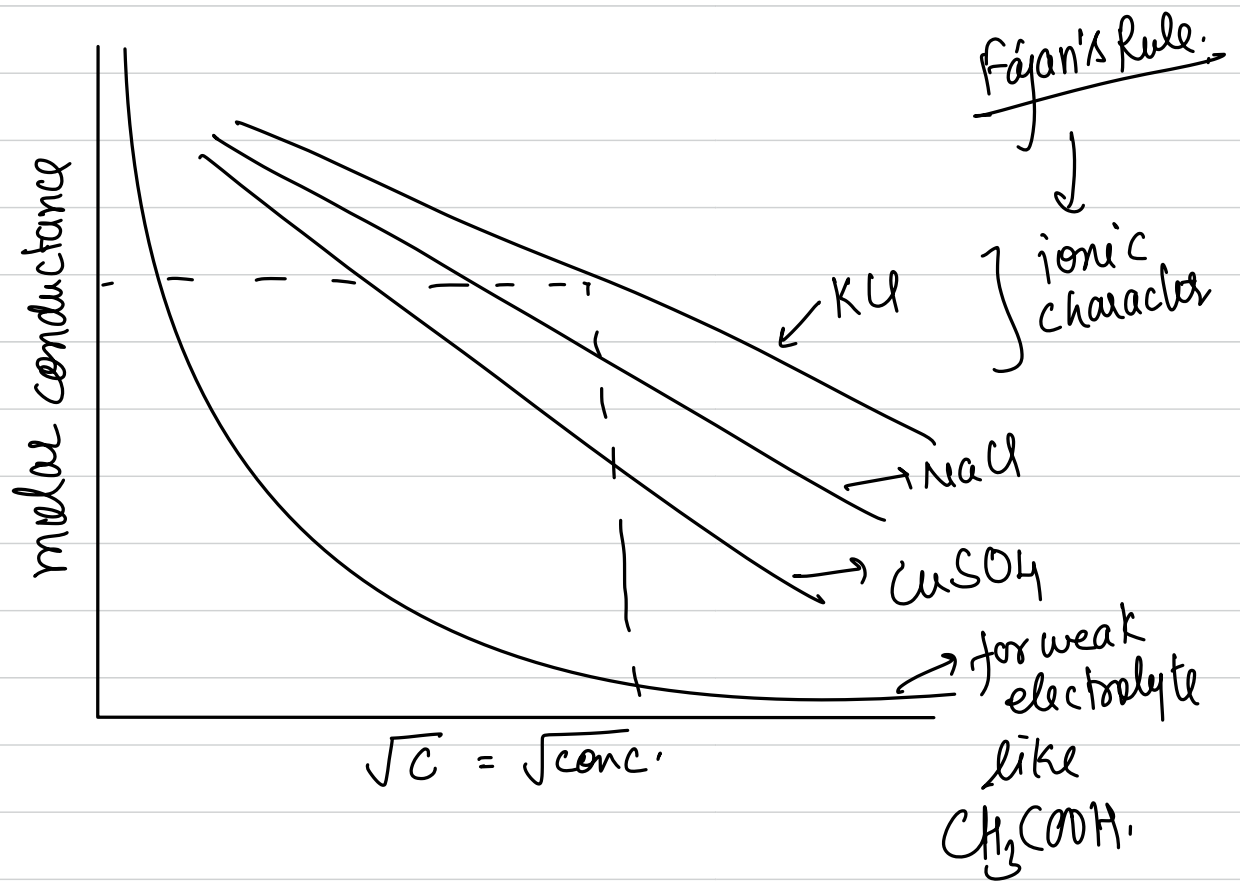
for strong electrolytes:-

ONSGER Equation:-

$$\Lambda_m = \Lambda_m^{\circ} - A\sqrt{c}$$

c = conc. of electrolyte

$A \rightarrow$ const. for a given solvent & temp. it only depends upon the type of electrolyte.



Calculation of degree of dissociation for weak electrolyte

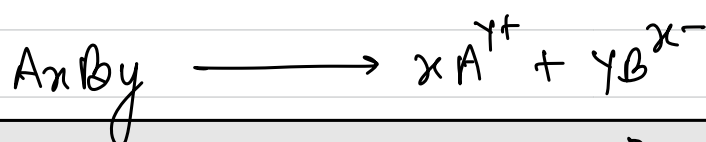
$$\alpha = \frac{\Lambda_m}{\Lambda_m^\infty}$$

→ conductance will be max.

Kohlrausch's law:-

At infinite dilution, when the dissociation is complete, each ion makes a definite contribution towards molar conductance of the electrolyte irrespective of the nature of the other ion with which it is associated & the molar conductance at infinite dilution for any electrolyte is given by contributions of two ions

$$\Lambda_m^{\circ} = \lambda_{+}^{\circ} + \lambda_{-}^{\circ}$$



$$\Lambda_m^{\circ} = x \lambda_{A^{y+}}^{\circ} + y \lambda_{B^{x-}}^{\circ}$$

Q: Calculate the value of Λ_m° for NH_4OH from the following given data

$$\Lambda_m^\circ \text{ for } \text{Ba}(\text{OH})_2 = 457 \, \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$$

$$\Lambda_m^\circ \text{ for } \text{BaCl}_2 = 240.6 \, \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$$

$$\Lambda_m^\circ \text{ for } \text{NH}_4\text{Cl} = 129.8 \, \Omega^{-1} \text{cm}^2 \text{mol}^{-1}.$$

$$\Lambda_m^\circ \text{ for } \text{NH}_4\text{OH} = 238 \, \underline{\underline{\Omega^{-1} \text{cm}^2 \text{mol}^{-1}}}$$

Batteries

- * Batteries have one or more than one cell connected in series.
- * Function of batteries is to convert the chemical energy into electrical energy.
- * The battery should be light, compact & its voltage should not vary appreciably during its use.

Types of Batteries

Primary

Secondary

Primary Batteries:-

The reaction occurs only once and after use over a period of time the battery becomes dead over a period of time and can not be used again.

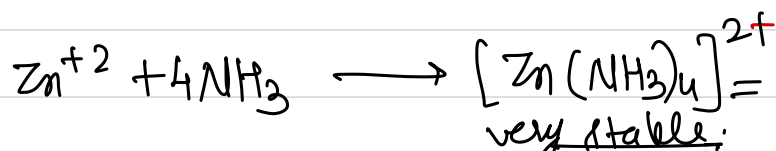
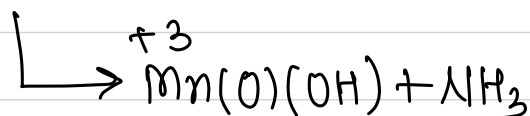
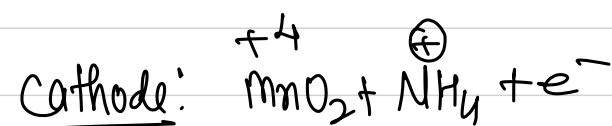
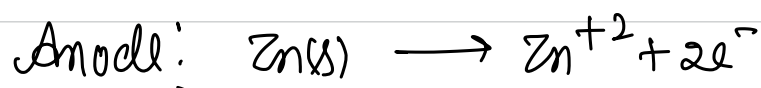
- * The most familiar example of this type is dry cell (or Leclanche cell) which is used in clocks & transistors (Penul cell).

cell contains →

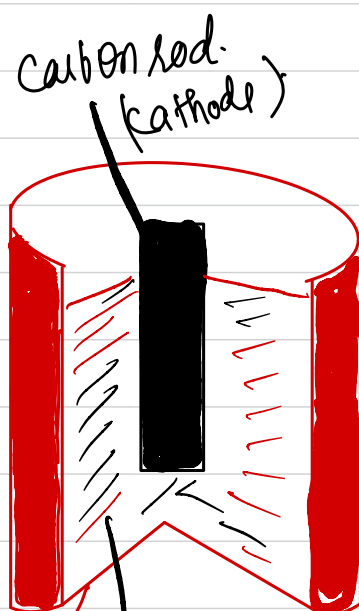
cell container acts as an anode and is made up of Zn.

Cathode is made up of graphite surrounded by powdered MnO_2 & C.

* space b/w the electrodes is filled by a moist paste of NH_4Cl & ZnCl_2 .



cell potential = 1.5 V



MnO_2 + carbon-black.
+ NH_4Cl paste

Zinc cup (anode)

Mercury cell:-

This type of cell is used in low current devices like hearing aid & watches.

It consist of Zn-Hg as anode & a part of HgO &c. act as cathode.

The electrolyte is a paste of KOH + ZnO.

Anode:-



Cathode:-



cell potential = 1.35V.

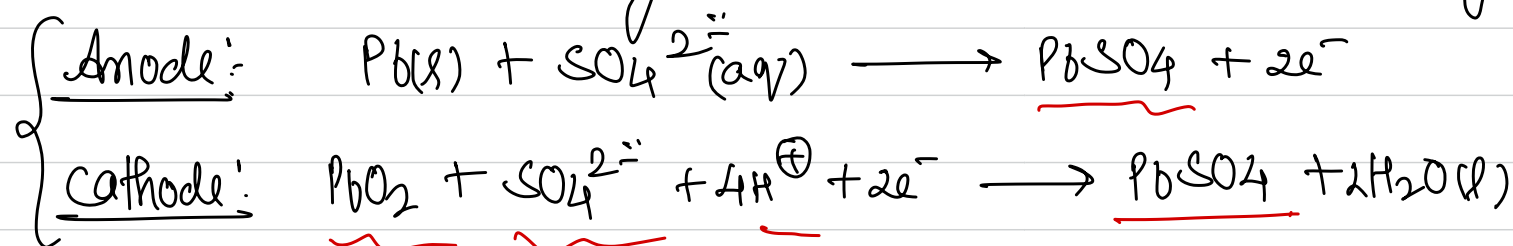
Secondary Batteries (Rechargeable Batteries):-

A good secondary cell can undergo a large no of discharging & charging cycles.

The most important secondary cell is lead storage batteries.

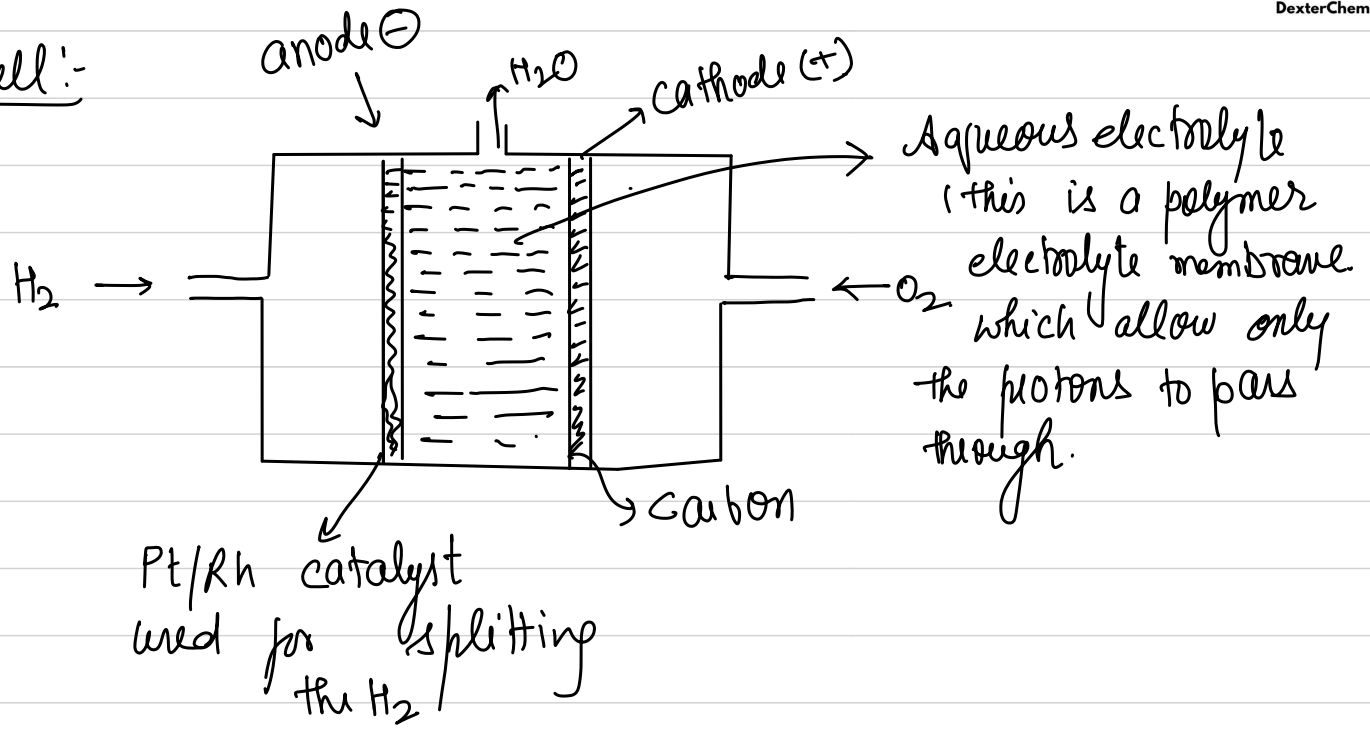
* Lead anode & grid of lead packed with PbO_2 act as cathode.

* 38% solution of H_2SO_4 is used as an electrolyte.



these rxns are for discharging of the Battery
 On charging the battery, rxns are reversed
 & $PbSO_4$ on anode & cathode is converted into
 $Pb(s)$ & PbO_2 respectively

Fuel cell:-



Anode:



Cathode:



Conductometric Titration :-

Conduction of ions depends upon.

① Charge :- Greater the charge better will be the conduction.

② size :- (hydrated state) :-

smaller the cation greater will be conduction.

③ Hydration :-

extent of hydration $\propto \frac{1}{\text{conductivity}}$

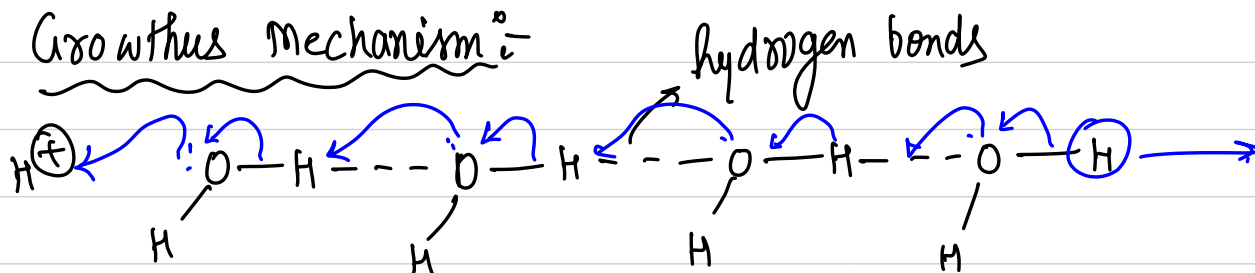
hydration $\propto \frac{1}{\text{ionic mobility}}$

hydration $\propto \frac{1}{\text{size}}$ (smaller the size greater will be hydration)

hydration $\propto \text{charge}$

* The highest conductivity is of H^+ ions & then OH^- ions & then follows the rest.

Gronthaus Mechanism :-



PCET $\rightarrow$ proton coupled electron transfer.
electron coupled proton transfer

Titration curves:-

① Strong acid HCl + strong Base NaOH

